

Ten Principles of Economics & Thinking Like an Economist

EC107: Microeconomics

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Summer 2026

Where we are going today

- ➊ What economics is about
- ➋ How people make decisions (1–4)
- ➌ How people interact (5–7)
- ➍ How the economy works (8–10)
- ➎ Thinking like an economist
- ➏ Independence & Trade
- ➐ Production possibilities
- ➑ Comparative advantage
- ➒ The gains from trade
- ➓ Terms of trade
- ➑ From nations to individuals
- ➒ Sources, winners and losers
- ➓ Wrap-up

One word: scarcity

Definition

Economics is the study of how society manages its *scarce* resources.

Scarcity = resources are limited, but wants are not.

- Nobody can have everything they want $\Rightarrow$ choices must be made.
- Every choice has a cost: what you gave up to get it.

Everyday scarcity

You have 24 hours and an exam in two days. Sleep, revise, or see friends? There is no option where you do all three fully. *That* is the economic problem.

The plan: 10 principles, 3 groups

How people *decide*

1. Trade-offs
2. Opportunity cost
3. Thinking at the margin
4. Incentives

How people *interact*

5. Trade can help all
6. Markets organise activity
7. Governments can improve outcomes

How the *economy* works

8. Productivity → living standards
9. Money growth → inflation
10. Short-run inflation/unemployment trade-off

Principle 1 — People face trade-offs

“There is no such thing as a free lunch.”

- To get one thing we usually give up another.
- Classic macro trade-off: *guns vs. butter* — defence spending vs. consumer goods.
- A society-level trade-off you will meet all term: **efficiency vs. equality**.

Efficiency vs. equality

Efficiency: getting the most out of scarce resources (size of the pie).

Equality: distributing it fairly (how the pie is sliced).

Policies that redistribute (welfare, progressive tax) can shrink the pie by blunting incentives. Recognising the trade-off does not tell us what to choose — but ignoring it leads to bad policy.

Principle 2 — The cost of something is what you give up

Opportunity cost

The **opportunity cost** of an item is what you give up to get it — the value of the *next-best* alternative forgone.

What does a year at university actually cost?

Looks like:

- Tuition: £9,250
- Books & living: £8,000

Economist adds:

- Forgone salary (**not** working): £22,000

Rent and food are *not* full opportunity costs — you would pay to eat and sleep anyway. But the salary you don't earn is the big hidden cost.

Takeaway

The true cost is dominated by the time, not the tuition. This is why pro athletes often skip university — their forgone salary is enormous.

Principle 3 — Rational people think at the margin

Marginal change

A small incremental adjustment to a plan of action. A rational decision-maker acts if and only if

$$\text{marginal benefit} > \text{marginal cost}.$$

The airline standby seat. A flight costs £100,000 to operate; 200 seats $\Rightarrow$ average cost £500/seat. A standby passenger offers £300. Refuse? *No*: the plane is flying anyway, so the **marginal** cost of one more passenger (a soda and a bag of peanuts) is near zero. £300 > £0. Take it.

The diamond–water paradox

Water is essential, diamonds are not — yet diamonds cost far more. Why? Because price reflects *marginal* value. Water is abundant, so one more litre is worth little; diamonds are scarce, so one more is worth a lot.

Principle 4 — People respond to incentives

If marginal benefit or cost changes, behaviour changes. Good policy anticipates this; bad policy forgets it.

Seat belts and the Peltzman effect (1975)

Mandatory seat belts make crashes more survivable $\Rightarrow$ drivers feel safer $\Rightarrow$ some drive a little faster/less carefully. Result: fewer deaths *per crash*, but **more crashes** — and more pedestrian deaths. Not an argument against seat belts; an argument for taking incentives seriously.

Unintended consequences are everywhere

- A tax on plastic bags $\Rightarrow$ sharp drop in bag use.
- “Cobra bounty” in colonial Delhi $\Rightarrow$ people *bred* cobras for the reward.

Your turn: spot the principle

Your turn

For each, name the principle (1–4):

- ❶ “I studied an extra hour because the exam is worth 40%.”
- ❷ “Going to the gig means missing my friend’s dinner.”
- ❸ “The all-you-can-eat buffet made me eat the 4th plate even though I was full.”
- ❹ “Higher carbon taxes pushed the firm toward solar.”

(Answers: 3 margin / 4 incentive. 2 opportunity cost / 1 trade-off; 3 margin again — MB falling; 4 incentives.)

Principle 5 — Trade can make everyone better off

- Trade is **not** a contest with a winner and a loser.
- It lets people *specialise* in what they do best and buy the rest cheaply.
- Countries gain the same way — this is the heart of **Day 2**.

Think locally

Your family doesn't sew its own clothes, grow its own wheat, and build its own phone. You specialise (a job), earn income, and trade for everything else — and you're richer for it. A nation is the same idea, scaled up.

Principle 6 — Markets are usually a good way to organise activity

The invisible hand (Adam Smith, 1776)

Self-interested buyers and sellers, guided by **prices**, are led “as if by an invisible hand” to outcomes that are often good for society as a whole — *without* anyone planning it.

Prices are signals. They carry information about scarcity and value:

- A frost in Brazil $\Rightarrow$ coffee price rises $\Rightarrow$ consumers economise, suppliers ramp up — automatically.

Why central planning struggled

The USSR set prices by command. With no signal of true scarcity, factories made millions of left shoes and shortages of bread coexisted with surpluses elsewhere. Prices weren't allowed to do their job.

Principle 7 — Governments can sometimes improve market outcomes

Markets work *usually*, not *always*. Government has two jobs:

Promote efficiency

Fix **market failures**:

- *Externalities* — pollution affecting bystanders.
- *Market power* — a monopolist distorting prices.

Promote equality

- Redistribute via taxes & transfers.
- The efficiency–equality trade-off (Principle 1) returns here.

Also foundational: **property rights** and the **rule of law**. Nobody invests if the harvest can be seized. “Can” is doing real work — government *can* improve outcomes, but doesn’t always.

Principle 8 — Living standards depend on productivity

Productivity

The quantity of goods and services produced per hour of a worker's time.

Almost *all* differences in living standards — across countries and across time — trace back to differences in productivity.

The numbers

Average income in the US/UK is roughly 8–10× that of low-income countries, and today's worker is far richer than their 1900 counterpart. Not because they work harder — because each hour produces vastly more (education, capital, technology).

Implication for policy: to raise wages sustainably, raise productivity — not by decree, but through skills, investment, and innovation.

Principle 9 — Prices rise when too much money is printed

Inflation

A sustained increase in the overall price level.

Persistent, high inflation almost always has one cause: **growth in the quantity of money**. When a government prints money to pay its bills, each unit buys less.

Hyperinflation case files

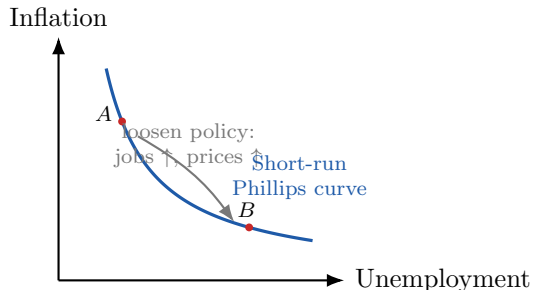
- *Weimar Germany, 1923*: prices doubled every few days; people burned banknotes for heat because they were cheaper than firewood.
- *Zimbabwe, 2008*: a 100-trillion-dollar note was issued. Bread cost billions by afternoon.

Principle 10 — Short-run trade-off: inflation vs. unemployment

In the *short run*, policies that pump money into the economy tend to:

- raise spending and demand;
- raise prices (inflation $\uparrow$);
- firms hire more (unemployment $\downarrow$).

This trade-off (the **Phillips curve**) is temporary, but it is why managing the *business cycle* is hard: cooling inflation can cost jobs.



The ten principles at a glance

Decisions

1. People face trade-offs
2. Cost = what you give up
3. Rational people think at the margin
4. People respond to incentives

Interaction

5. Trade can benefit everyone
6. Markets are usually good organisers
7. Government can sometimes help

The economy as a whole

8. Productivity determines living standards
9. Money growth causes inflation
10. Short-run inflation/unemployment trade-off

Keep these in your pocket

Almost every result this term is one of these ten ideas, made precise.

The economist as scientist

Economists try to be *objective*: observe, theorise, test.

The scientific method, applied to society

Observe → build a theory → gather data → test and revise.

The hard part: we usually **cannot run experiments**. You can't give one country hyperinflation just to see what happens. So economists exploit *natural experiments* — history's accidents (a sudden border change, a policy that hit one region not another).

Example

Germany's post-war division gave a near-experiment: same people, same culture, two economic systems. The outcome (West vs. East by 1989) is hard to ignore.

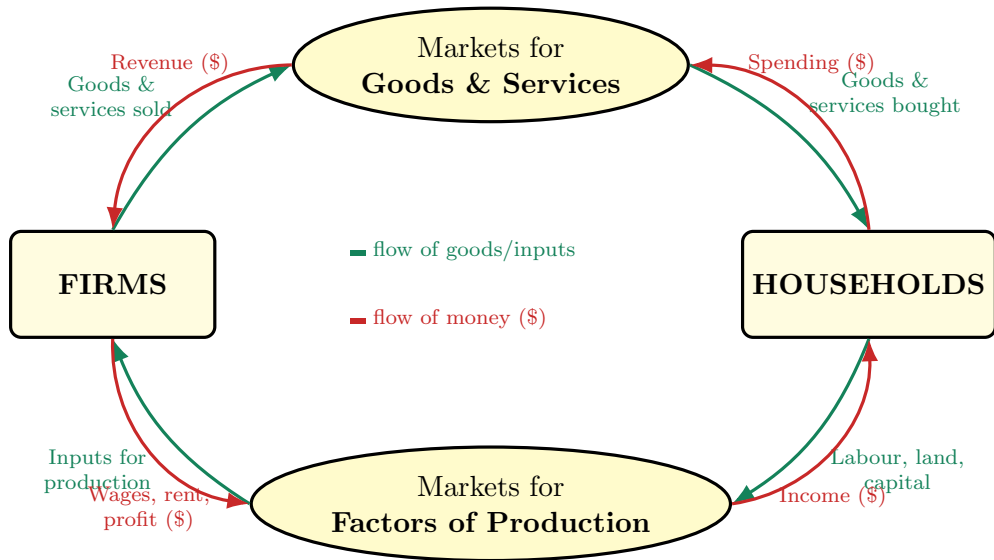
The role of assumptions and models

- Good assumptions **simplify** without distorting the question at hand.
- Physicists assume no air resistance to study a falling marble — fine for a marble, useless for a feather. Economists choose assumptions the same way.
- A **model** is a deliberate simplification — a map, not the territory. A useful map leaves things out on purpose.

We will use two foundational models today:

- ① The **circular-flow diagram**.
- ② The **production possibilities frontier (PPF)**.

Model 1: The circular-flow diagram



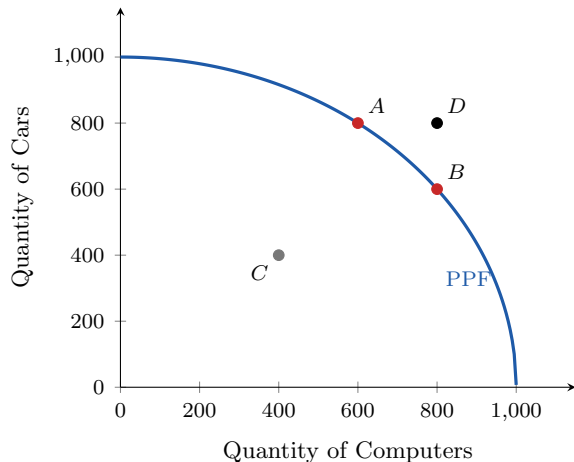
Reading the circular flow

- **Households** own the factors (labour, land, capital) and consume goods.
- **Firms** buy factors, produce, and sell goods.
- In the *goods market*: firms sell, households buy.
- In the *factor market*: households sell (their labour), firms buy.
- For every **green** arrow (a real good) there is an opposite **red** arrow (a payment).

What it teaches

The economy is a closed loop: one person's spending is another's income. Your wage is a firm's cost; your purchase is a firm's revenue.

Model 2: The production possibilities frontier



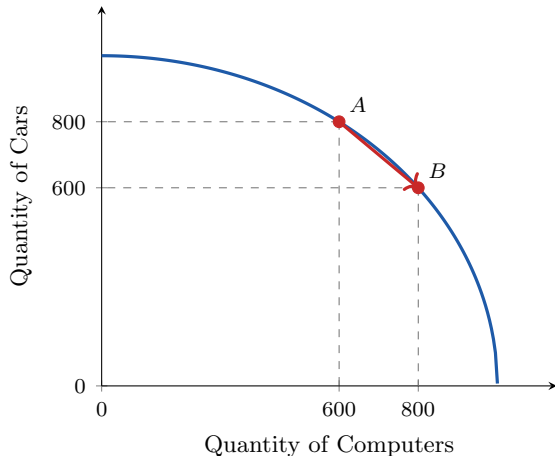
The PPF shows the **combinations of two goods** the economy can produce with its resources and technology.

- **A, B**: **on** the curve — *efficient* (no waste).
- **C**: **inside** — *inefficient* (idle resources, e.g. a recession).
- **D**: **outside** — *infeasible* with current resources.

Bowed-out shape

Resources aren't equally good at both jobs. Shifting them gets progressively more costly $\Rightarrow$ the curve bows outward.

The PPF shows opportunity cost



Move from *A* to *B*:

- Computers: 600 $\rightarrow$ 800 (+200)
- Cars: 800 $\rightarrow$ 600 (−200)

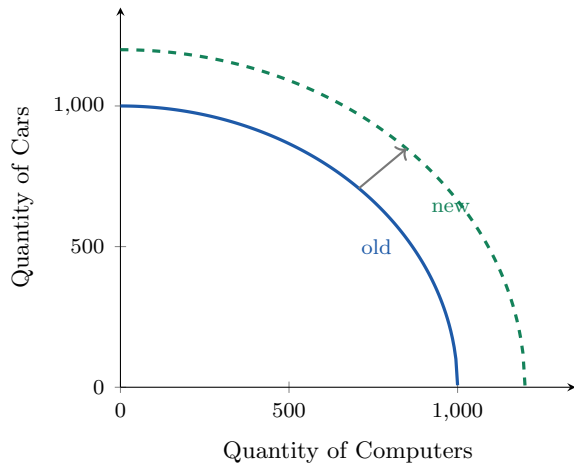
Opportunity cost

To make 200 more computers, the economy gives up 200 cars.

Opportunity cost of 1 computer $\approx$ 1 car *here*.

The **slope** of the PPF *is* the opportunity cost — and because the curve bows out, that cost rises as you specialise further.

Growth shifts the PPF outward



A technological advance or more resources lets the economy make *more of both* goods.

- The frontier shifts **outward** (green dashed).
- Points that were infeasible (like *D* earlier) become reachable.

This is Principle 8

Economic growth = pushing the PPF out, driven by **productivity**.

Two branches of economics

Microeconomics (*this course*)

How *individual* households and firms decide and interact in markets.

- Why do flight prices change?
- How does a tax affect buyers vs. sellers?

They are linked: macro outcomes are built from millions of micro decisions — but each needs its own tools.

Macroeconomics

Economy-wide phenomena.

- Inflation, unemployment, growth.
- Central-bank interest rates.

Positive vs. normative statements

Positive (*is*)

Descriptive claims about how the world *is*. Can be confirmed or refuted by data.

- “A minimum wage *raises* unemployment among low-skilled workers.”

Normative (*ought*)

Prescriptive claims about how the world *ought* to be. Involve *values*.

- “The government *should* raise the minimum wage.”

Why it matters

Data can settle a positive dispute. A normative dispute needs *values* too — no amount of data alone decides what we *should* do.

Why do economists disagree?

- ① **Different scientific judgements** — they hold competing positive theories about how the world works (e.g. how responsive is saving to tax cuts?).
- ② **Different values** — normative disagreements about goals (efficiency vs. equality again).

They agree more than headlines suggest

Large majorities of economists agree that, e.g., *rent control reduces the quantity and quality of housing* and *tariffs usually reduce general welfare*. Disagreement on TV is louder than disagreement in the data.

Key takeaways - 1

- Economics studies choices under **scarcity**.
- Decisions: *trade-offs, opportunity cost, the margin, incentives*.
- Interaction: *trade gains, markets & the invisible hand, a role for government*.
- The economy: *productivity, money & inflation, the short-run trade-off*.
- Economists think with **models** — the circular flow and the PPF.
- Keep *positive* and *normative* claims apart.

Coming next — Day 2

Interdependence and the Gains from Trade: we make Principle 5 precise with **absolute** and **comparative advantage**. Why even a worse-at-everything country still gains from trade.

INDEPENDENCE & TRADE

Samuelson's challenge

The mathematician Stanislaw Ulam once teased the economist Paul Samuelson: name one proposition in all of the social sciences that is both **true** and **non-trivial**.

Samuelson's answer, years later

Comparative advantage. “That it is logically true need not be argued before a mathematician; that it is not trivial is attested by the thousands of important and intelligent people who have never been able to grasp the doctrine for themselves, or to believe it after it was explained to them.”

Today we will prove something that sounds impossible:

*A country that is worse at producing **everything** can still gain from trade — and so can the country that is better at everything.*

A thought experiment: the self-sufficient life

Imagine you had to consume only what *you* produced. No shops, no trade.

- Grow your own wheat, grind your own flour, build your own oven.
- Sew your own clothes from fibre you grew. Generate your own electricity.

A full day's brutal labour might buy you one bad meal and a draughty hut.

The point

Your high standard of living is *entirely* a product of **not** being self-sufficient — you specialise in one narrow thing and trade for everything else. Now scale this up: a country that trades with no one — economists call it **autarky** — falls into the same trap, magnified across millions of people.

Our laboratory: a two-country economy

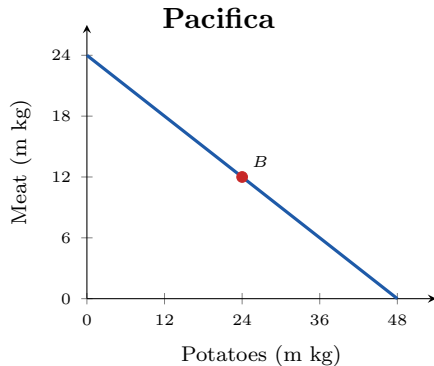
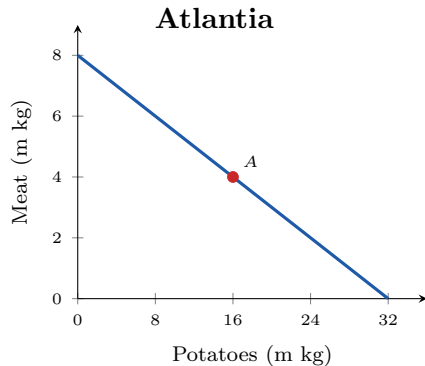
Two countries, **Atlantia** and **Pacifica**, each able to produce *meat* and *potatoes*. Devoting *all* its resources to a single good, each can make at most:

	Max meat	Max potatoes
Atlantia	8	32
Pacifica	24	48

- **Pacifica can out-produce Atlantia in *both* goods.** The naive conclusion — “so Pacifica should make everything” — is wrong.
- Each country’s *opportunity costs* are hidden in this table. We extract them next.

Quantities are in millions of kg per year — the model is stylised; the logic is exact.

Each country's PPF, and life without trade



The PPFs are *straight lines*, so each country's opportunity cost is **constant**. Points *A*, *B* are what each produces and consumes alone — here, splitting resources evenly.

The tempting wrong answer: absolute advantage

Absolute advantage

Producing a good with *fewer resources* than another producer. Pacifica has absolute advantage in **both** meat and potatoes — it can make more of each.

So shouldn't Pacifica just make everything and Atlantia produce nothing worth trading?

Why that's wrong

Pacifica's resources are *finite*. Every factory and field used for potatoes is one *not* used for meat. The right question is never “who is more productive?” but “what does each country *give up*?” — the **opportunity cost**.

This is Principle 1 from yesterday (*there are no free lunches*) doing real analytical work.

Opportunity cost and comparative advantage

Read each opportunity cost straight off the PPF (max of one good $\div$ max of the other):

	Cost of 1 kg meat	Cost of 1 kg potatoes
Atlantia	4 kg potatoes	1/4 kg meat
Pacifica	2 kg potatoes	1/2 kg meat

Comparative advantage

Producing a good at a *lower opportunity cost* than another producer.

Meat: Pacifica gives up 2 kg potatoes, Atlantia 4.
 $\Rightarrow$ **Pacifica.**

Potatoes: Atlantia gives up 1/4 kg meat, Pacifica 1/2.
 $\Rightarrow$ **Atlantia.**

The key theorem

Comparative advantage is always *split*

One country cannot have a comparative advantage in *both* goods. If your opportunity cost of meat is lower, your opportunity cost of potatoes is mechanically *higher* — they are reciprocals. So there are always gains to divide, no matter how lopsided absolute ability is.

- Absolute advantage can be one-sided (Pacifica has both). *Comparative* advantage never is.
- This is why even a country that is worse at everything **always** has something to export: the good it is *least bad* at.

Trade flows follow *comparative*, never absolute, advantage — Ricardo's 1817 insight.

Specialise, then trade

Each country specialises toward its comparative advantage:

- **Atlantia** → all potatoes: produces (32 potatoes, 0 meat).
- **Pacifica** → mostly meat: produces (12 potatoes, 18 meat).

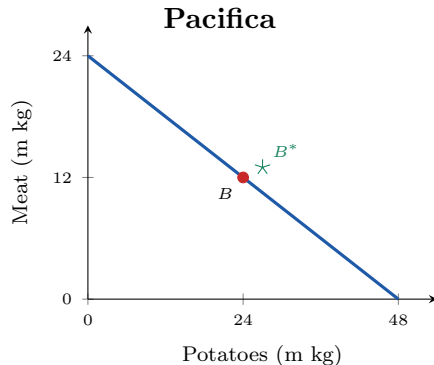
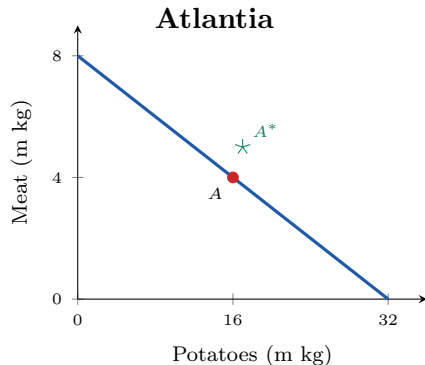
The proposed deal

Atlantia exports **15 kg potatoes** in return for **5 kg meat** from Pacifica.

Implied price: 1 kg meat = 3 kg potatoes.

We now check: does this leave *both* countries better off than going it alone? (It must, or one would refuse.)

Both consume beyond their own PPF



The green star (A^* , B^*) is consumption *with* trade — it lies **outside** the country's own PPF. No amount of self-sufficiency could reach it.

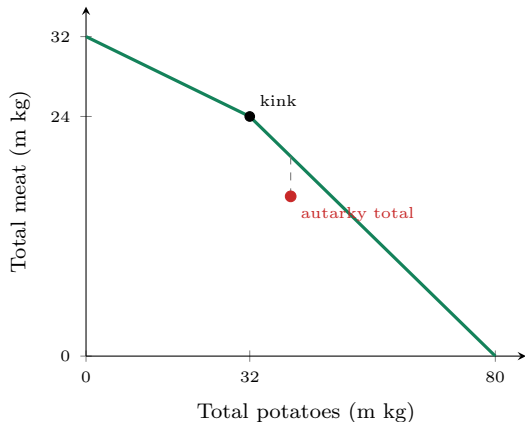
The gains, quantified

		Meat	Potatoes
Atlantia	alone (A)	4	16
	with trade (A^*)	5	17
	gain	+1	+1
Pacifica	alone (B)	12	24
	with trade (B^*)	13	27
	gain	+1	+3

Where did the extra food come from?

No country worked longer; no new land or livestock appeared. Trade did not add *resources* — it reallocated *production* to whoever sacrifices least. Specialisation by comparative advantage is a genuine free lunch at the level of the world economy.

The whole world moves to its frontier



The world PPF (summing both countries) is **bowed out** with a *kink*, even though each country's PPF is straight.

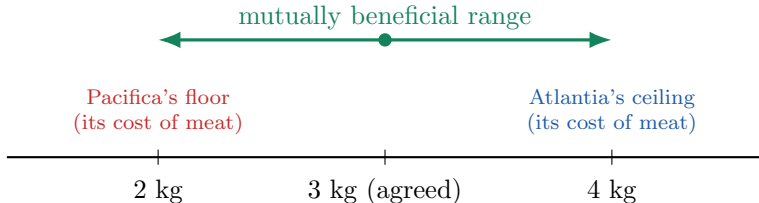
- To make meat, the world uses the **low-cost** meat producer first — Pacifica.
- Only once Pacifica is fully specialised does Atlantia switch into meat.
- The kink is exactly *full specialisation by comparative advantage*.

Autarky totals sit *inside* the frontier; specialisation captures the gap.

What price will they agree? Deriving the range

For *each* country to gain, the trade price must beat its own do-it-yourself option.

- **Pacifica** exports meat. It won't accept fewer than **2 kg potatoes** per kg of meat — that is what meat costs *it* to make. (Below 2: it would rather grow potatoes itself.)
- **Atlantia** imports meat. It won't pay more than **4 kg potatoes** per kg of meat — that is what meat costs *it*. (Above 4: it would rather make its own.)



Where in (2, 4) the price lands is set by world supply and demand for the two goods.

Your turn — find the comparative advantage

Your turn

Country X can produce 10 cars or 40 tonnes of grain per period. Country Y can produce 2 cars or 20 tonnes of grain per period.

- ① Who has the *absolute* advantage in each good?
- ② State each country's opportunity cost of *one car*.
- ③ Who should specialise in cars, and what range of “grain-per-car” prices makes trade worthwhile for both?

(1) X, in both. (2) X: 1 car = 4 tonnes grain; Y: 1 car = 10 tonnes grain. (3) X specialises in cars (lower cost). Any price between 4 and 10 tonnes of grain per car benefits both.

The same logic runs at every scale

We proved it for countries — but nothing in the argument used the word “country”. The identical reasoning governs **individuals**.

The surgeon and the typist

A surgeon happens to be the fastest typist in the hospital too — absolute advantage in both. Should she type her own notes?

No. An hour she spends typing is an hour *not* operating — an enormous opportunity cost. The typist’s opportunity cost of an hour is tiny. Both gain when she operates and pays the typist to type, *even though she is better at both*.

Comparative advantage runs seamlessly from **individuals** to **firms** to **nations** — it is the reason for specialisation everywhere.

Where does comparative advantage come from?

- **Natural resources & climate.** Saudi oil, Brazilian coffee, Icelandic geothermal.
- **Factor endowments.** Labour-abundant countries lean to labour-intensive goods; capital-abundant ones to capital-intensive goods. (*Heckscher–Ohlin.*)
- **Technology & know-how.** German precision engineering, Swiss finance, Silicon Valley software.
- **Accumulated skill and scale.** Often *built*, not inherited — comparative advantage can be **dynamic**, which is what industrial policy tries (riskily) to exploit.

Implication

A country's trade pattern is not destiny written by geography alone — but it is heavily shaped by what it is relatively well-equipped to do *today*.

The part most cheerleaders skip: winners *and* losers

Trade raises a nation's **total** income. It does *not* make every individual better off.

Why

Opening to trade is, for the import-competing industry, like a fall in the price of what they sell. Workers and owners there can be made *worse off*, even as consumers and exporters gain more than enough to compensate them *in principle*.

- **Aggregate gain** (efficiency) and **distribution** (who gains) are different questions.
- “The winners could compensate the losers” is a statement about *potential*, not about what actually happens — and politically, the compensation often doesn't.

This is why trade is economically near-settled yet politically explosive.

Two real cases

The Corn Laws (Britain, repealed 1846)

Tariffs kept grain prices high, benefiting landowners at consumers' expense. Ricardo argued for repeal on comparative-advantage grounds; the fight pitted landlords against manufacturers and the urban poor. Repeal let Britain import cheap grain and specialise in manufactures — a textbook case of gains *and* of who resists them.

The “China shock” (roughly 1999–2011)

Rapid growth in Chinese manufacturing imports delivered cheaper goods to hundreds of millions of consumers — the aggregate gain. But the losses were *concentrated*: specific manufacturing regions in the US and Europe suffered lasting job losses and slow adjustment. Both halves of the theory showed up at once.

(Figures here are illustrative; I can pull precise, cited estimates on request.)

Common objections — and the economist's reply

Objection	Reply
“We can't compete with cheap foreign labour.”	Trade follows <i>comparative</i> , not absolute, cost. Low wages reflect low productivity; you still gain by specialising where your relative cost is lowest.
“A trade deficit means we're losing.”	An accounting identity, not a scoreboard: a current-account deficit is matched by a capital inflow. Imports are the <i>point</i> of exporting.
“If we're worse at everything, we have nothing to offer.”	Impossible. Comparative advantage is <i>relative</i> , so every country has one somewhere.
“Trade is one nation winning at another's expense.”	The central error. Voluntary trade is <i>positive-sum</i> — both sides consume beyond their PPF.

Where the simple model needs more (robustness)

Our model is a benchmark, not the whole story:

- **Adjustment is costly and slow.** Workers can't instantly retrain or relocate — the losers' pain is real and can persist.
- **Transport costs and barriers** shrink (sometimes erase) the gains.
- **Increasing returns & variety** (Krugman's "new trade theory") explain why similar rich countries trade similar goods (cars for cars) — driven by *scale* and *love of variety*, not comparative advantage.
- **Strategic, infant-industry and national-security** arguments are coherent in theory — and easily abused as cover for protectionism in practice.

The conclusion survives: trade is hugely beneficial in aggregate. The caveats are about *magnitude*, *speed* and *distribution*, not sign.

Misconceptions & exam traps

- **Using absolute advantage to assign production.** Always compare *opportunity costs*.
- **Thinking the stronger country shouldn't trade.** Pacifica gains too — specialisation frees its resources for what it sacrifices least.
- **Forgetting to check the price range.** A “trade” outside $(2, 4)$ here would make one country refuse. Beneficial trade needs a price between the two opportunity costs.
- **Equating aggregate gains with universal gains.** Trade is positive-sum for the *nation*; specific groups can still lose.
- **Reading a trade deficit as failure.** It is finance, not defeat.

Key takeaways - 2

- Interdependence exists because specialisation lets countries consume **beyond their PPF**.
- **Absolute advantage** (fewer resources) is a distraction; **comparative advantage** (lower opportunity cost) decides who produces what.
- Comparative advantage is *always* split — so gains from trade essentially always exist.
- Trade benefits both sides at any price *between* their opportunity costs.
- Aggregate gains are robust; **distribution** and **adjustment** are the real, legitimate complications.

Coming next

The Market Forces of Supply and Demand: how prices, including the terms of trade actually get set.

Questions?

EC107: Microeconomics - Week 1 Part 1